

Tennis Biomechanics - 3: The principle of “balance”



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This article presents the biomechanics principle of “balance” and its application in tennis. Balance is defined as the ability to manage body position, either statically or dynamically, in relation to one's base of support. Dynamic balance is involved in every steps, stops, stances, and moves. Faster is the game, higher are the demands in dynamic balance.

What are the factors affecting balance?

- Size of the base of support in the direction of force (neutral stance more stable than open stance for example)
- Height of the center of gravity above the base of support
- Location of the center of gravity projection within the base of support
- Mass of the body
- Friction related to the court surface (more stability on hard court than on clay court)

Practical coaching keys to improve balance

- Increasing the size of the base of support
- Lowering the center of gravity
- Keeping an upright position of head and upper body
- Using the non-dominant arm and leg to counteract the hip and shoulder rotations

Initial Lateral steps

Figures 1 and 2 highlight how Roger Federer maintains an upright position of the head and upper body with the lines of his eyes and his shoulders in horizontal position. In ready position (figures 1A and 2C), his feet are more than shoulder width apart with bent knees to lower his center of gravity.

Balance strategies during the groundstrokes:

Players use different actions of the feet and body to maintain dynamic balance when hitting the ball. They vary with the phase of the shot, the stance, the court position, the incoming ball's characteristics and their tactical intent.

Preparation phase

Figure 3 shows players at the end of the preparation phase of a forehand shot with a base of support larger than shoulder width apart both in neutral (Figure 3A) and open stance (Figure 3B). The non-dominant arm is positioned at the same height than the dominant arm and in the same plane.

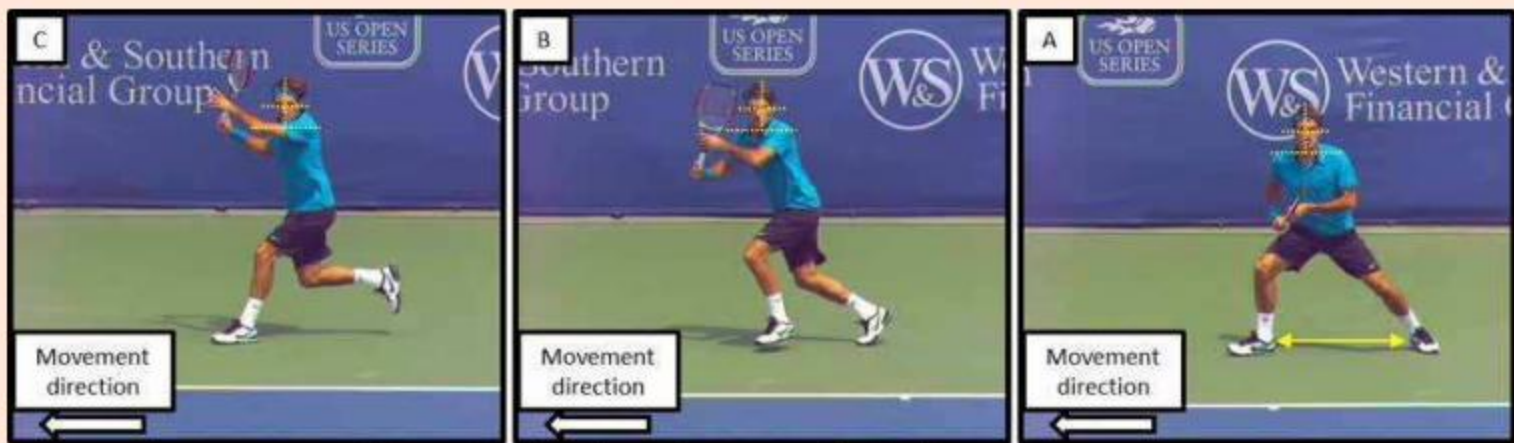


Figure 1: First lateral step to the ball



Figure 2: First lateral recovery step



Figure 3

Figure 4 shows players at the end of the preparation phase of a one-handed (figure 4A) and two-handed (figure 4B) backhand shot with a base of support larger than shoulder width apart

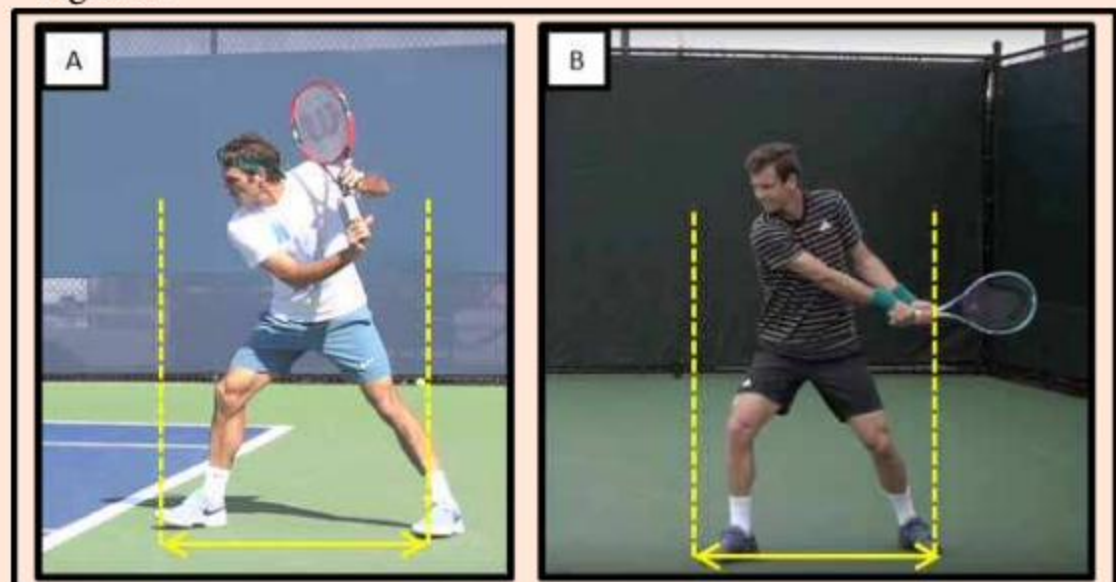


Figure 4

Acceleration phase

During the acceleration phase of groundstrokes, players use dynamic actions of non-dominant side limbs to maintain their balance depending on the stance used.

Figure 5 shows a player hitting a forehand with a neutral stance at knee (figure 5A) and waist (figure 5B) level. The player uses the back leg differently for balance depending on the height of the contact point. When the ball is low, the player drops the back knee down keeping it bent and close to the court. When the contact point is higher the player kicks the back leg back and up. The non-dominant arm is fixed to control the trunk rotation.

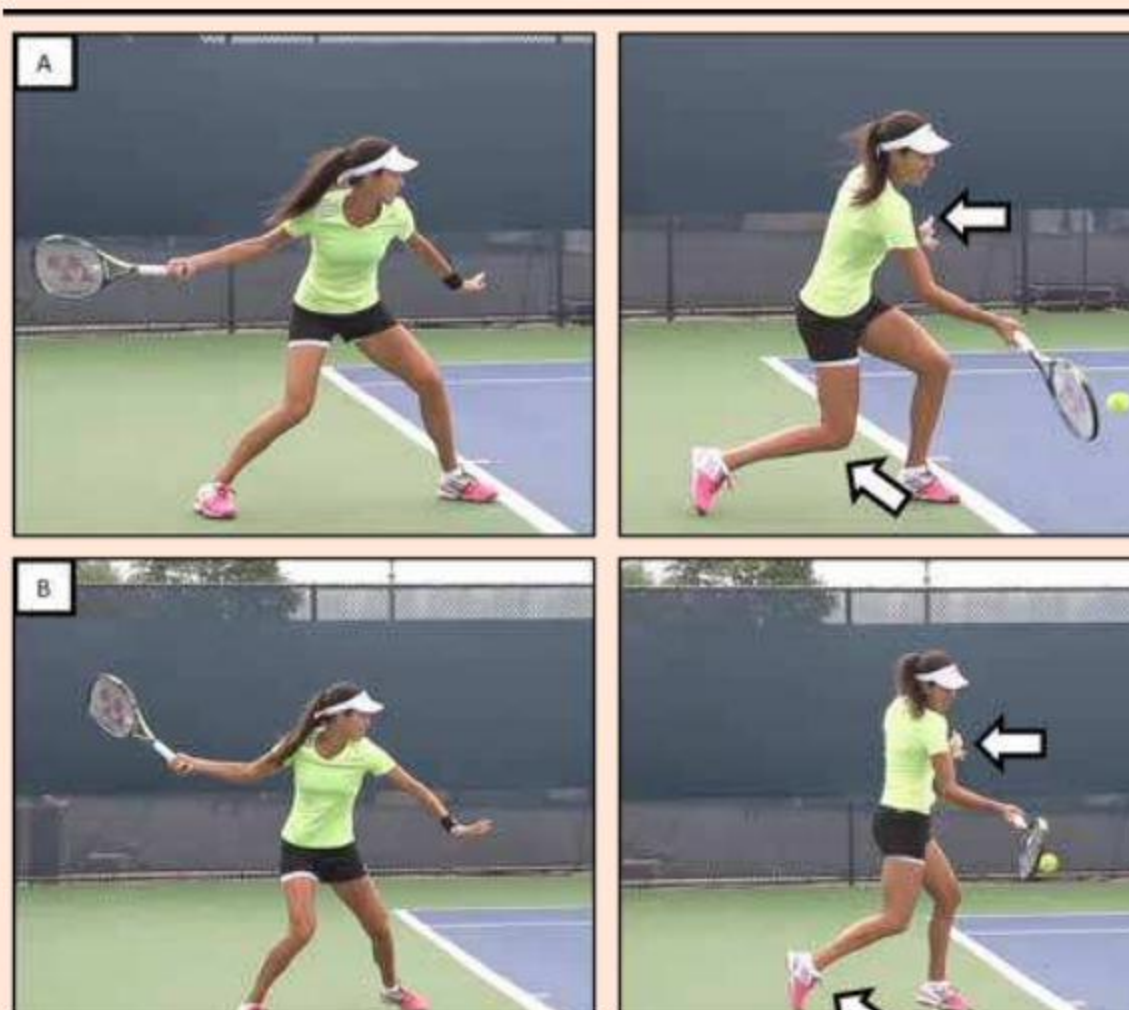


Figure 5

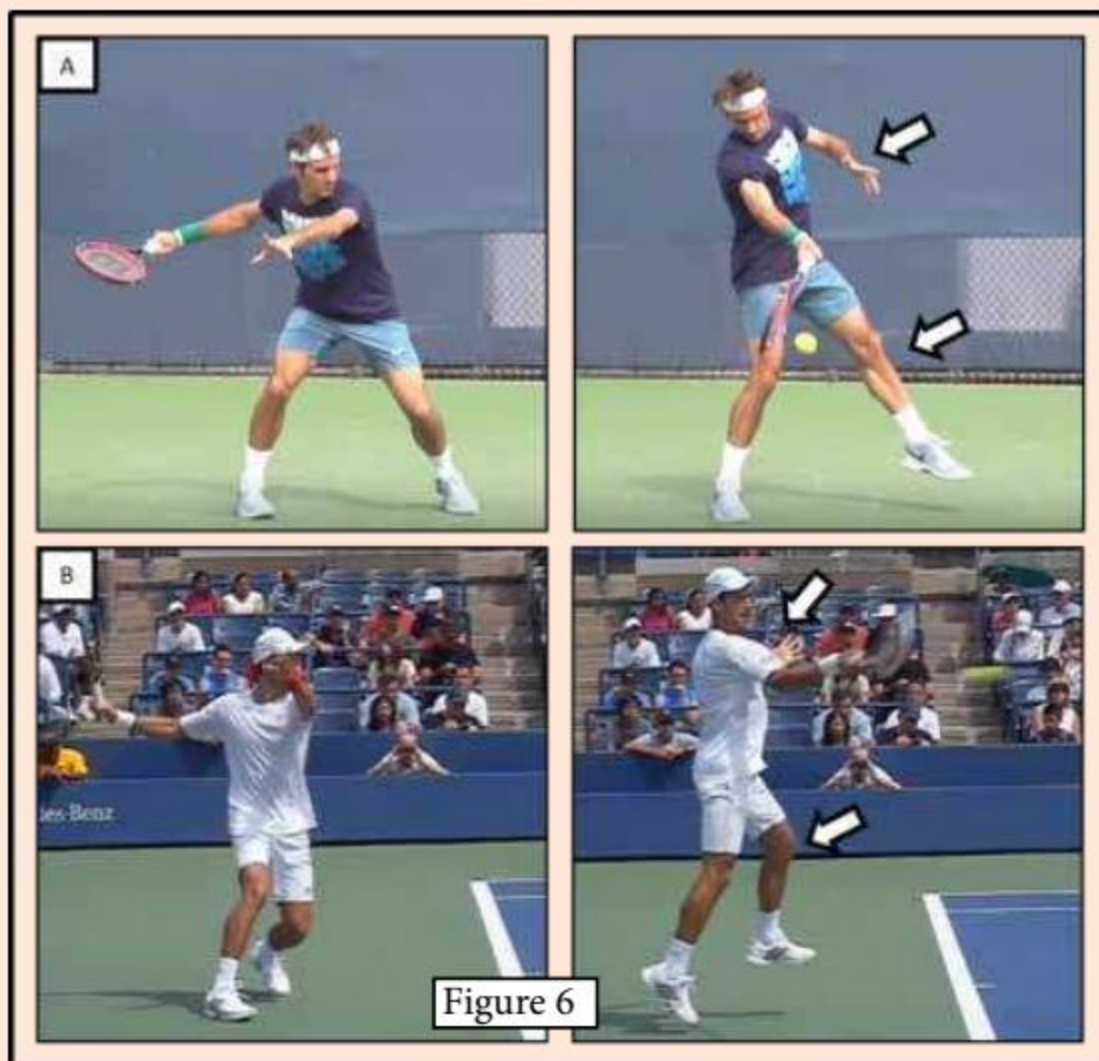


Figure 6

Figure 6 shows players hitting a forehand with an open stance under (figure 6A) and over (figure 6B) waist level. The players elevate the front leg and fix the non-dominant arm to counter the trunk rotation.

Balance strategies during the serve:

The foot technique affects the size of the base of support and thus the degree of stability (figure 7). Coaches have to monitor the balance ability of their players when asking them to perform a foot-up technique. From a development player point of view, the foot-back technique should be the first to be taught. Independently of the stance, the full non-dominant arm extension helps to stabilise the non-dominant side before initiating the leg drive.

During the acceleration and fol-

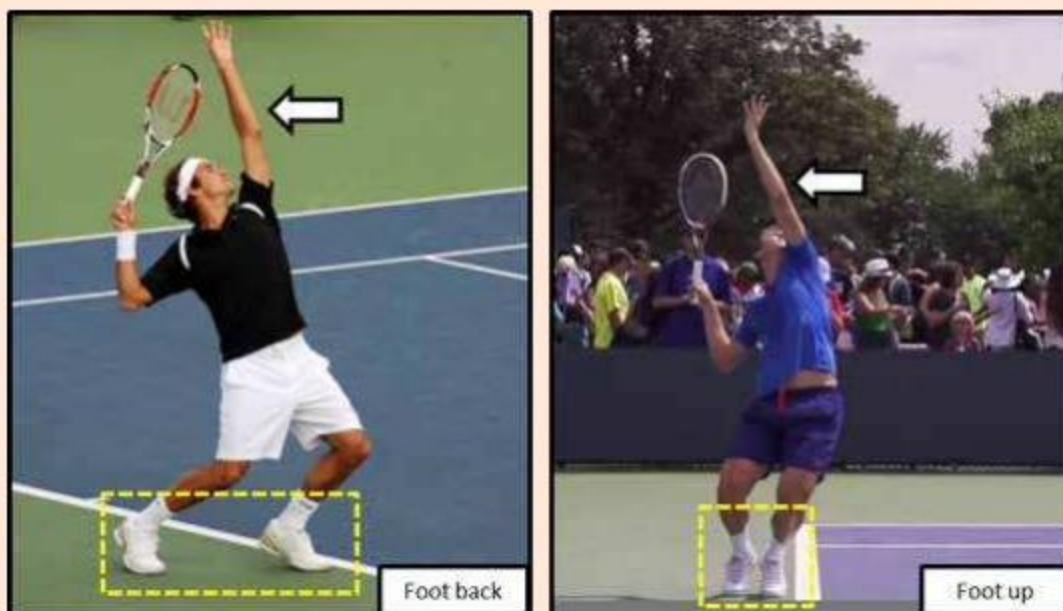


Figure 7



low-through phases, players use their non-dominant side to maintain dynamic balance:

- A full non-dominant arm extension allows vertical stabilization of non-dominant side (figure 8A).
- The non-dominant arm helps to block the upper body from excessive transverse rotation (Figure 8B)
- The Arabesc with the back leg combined with the backward extension of the non-dominant arm helps to counterbalance the forward flexion of the trunk at landing (figure 8C).

Balance strategies during the volley:

Figure 9 shows how the player uses the non-dominant arm and the non-supporting leg to maintain balance during

forehand (figure 9A) and backhand (figure 9B) volley.

Conclusion:

The “balance” principle is present in every tennis movement and shot production. Coaches should focus on ex-

ercises aiming at improve the ability of the player to be well-balanced. The exercises can be oriented both on technical and physical components depending on the player's needs.

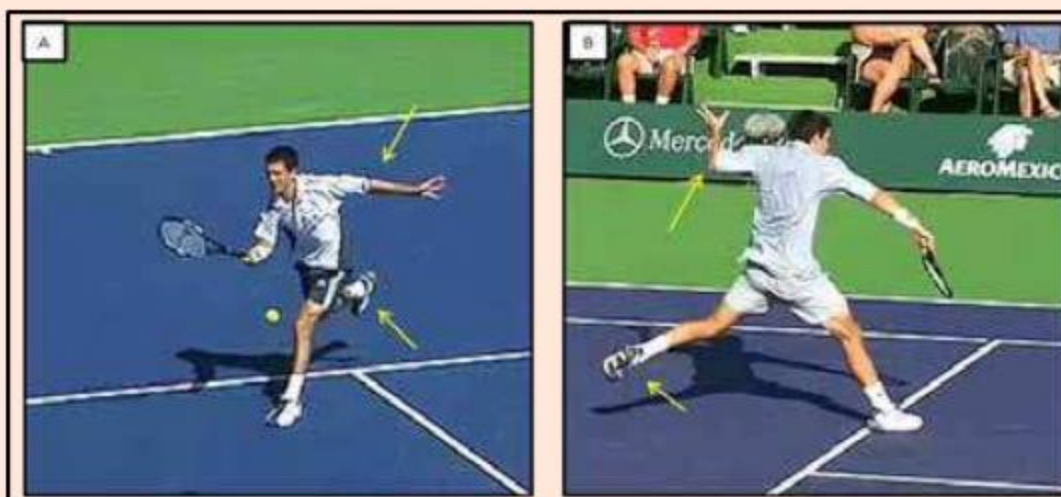


Figure 9